



SCEGGS Darlinghurst

**2004**

Higher School Certificate  
Trial Examination

FILE

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Centre Number

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Student Number

# Mathematics Extension 1

This is a TRIAL PAPER only and does not necessarily reflect the content or format of the Higher School Certificate Examination for this subject.

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## General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

## Total marks – 84

- Attempt Questions 1–7
- All questions are of equal value

Total marks – 84  
Attempt Questions 1–7  
All questions are of equal value

Answer each question on a NEW page

Question 1 (12 marks)

Marks

- (a) Solve for  $x$ :

3

$$\frac{3}{x-2} \leq 1$$

- (b) Find, to the nearest minute, the acute angle between the lines  $y = 4x + 5$  and  $3x + 2y - 1 = 0$ .

2

- (c) Find  $\lim_{x \rightarrow 0} \frac{\sin 4x}{8x}$

1

- (d) Evaluate  $\int_0^{\frac{\pi}{2}} \sin^2 3x \, dx$

3

- (e) Evaluate  $\int_0^1 x(1-x)^7 \, dx$  using the substitution  $u = 1 - x$ .

3

Question 2 (12 marks) START A NEW PAGE

Marks

- (a) Differentiate  $x^2 \sin^{-1} 3x$  with respect to  $x$ .

2

- (b) How many different arrangements of the letters of the word PARABOLA are possible?

2

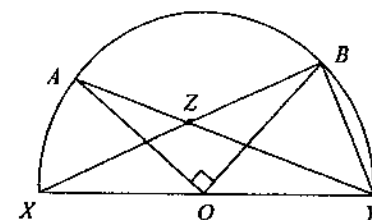
- (c) Find all real values of  $a$  for which  $P(x) = ax^3 - 8x^2 - 9$  is divisible by  $x - a$ .

2

- (d) The two curves  $y = \cos^{-1} x$  and  $y = 2 \tan^{-1}(1 - x)$  both cut the  $y$ -axis at the point  $\left(0, \frac{\pi}{2}\right)$ . Both curves also share a common tangent at  $\left(0, \frac{\pi}{2}\right)$ . Find the equation of this tangent.

2

- (e)



Not to scale

O is the centre of a semicircle, diameter XY.  
OA and OB are perpendicular, AY and XB intersect at Z.

Copy the diagram onto your answer sheet.

- (i) Explain why  $\angle AYB = 45^\circ$ .

1

- (ii) Prove that  $BY = BZ$ .

3

**Question 3 (12 marks) START A NEW PAGE**

**Marks**

- (a) (i) Express  $\sqrt{3} \cos x - \sin x$  in the form  $R \cos(x + \alpha)$  where  $R > 0$  and  $0 < \alpha < \frac{\pi}{2}$ . 2
- (ii) Hence, sketch the graph of the equation  $y = \sqrt{3} \cos x - \sin x$  for  $-\frac{\pi}{6} < x < 2\pi$ . 1
- (iii) Solve the equation  $\sqrt{3} \cos x - \sin x = \sqrt{2}$  for  $0 \leq x \leq 2\pi$ . 2
- (b) On a particularly windy day, a sock pegged on a clothes line is oscillating in simple harmonic motion such that its displacement,  $x$  centimetres, from the origin,  $O$ , is given by the equation:
- $$\ddot{x} = -16x \quad \text{where } t \text{ is the time in seconds.}$$
- (i) Show that  $x = a \cos(4t + \alpha)$ , where  $a$  and  $\alpha$  are constants, is a solution of motion for the sock. 1
- (ii) Initially, the sock is 5 cm to the right of the origin with a velocity of  $-4 \text{ cm s}^{-1}$ . Show that the amplitude of the oscillation is  $\sqrt{26}$  cm. 2
- (iii) Find the maximum speed of the sock. 1
- (c) Prove that  $5^n + 11$  is divisible by 4 for all integers  $n \geq 0$ , by mathematical induction. 3

**Question 4 (12 marks) START A NEW PAGE**

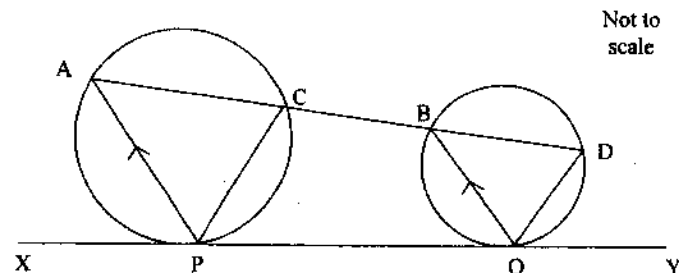
**Marks**

- (a) Consider the function  $f(x) = \pi + 2 \sin^{-1}\left(\frac{x}{3}\right)$
- (i) State the domain and range of  $y = f(x)$ . 2
- (ii) Sketch the graph of  $y = f(x)$ , marking clearly any endpoints. 2
- (b) Two roots of the equation  $x^3 + px^2 + q = 0$  ( $p, q$  real) are reciprocals of each other.
- (i) Show that the third root is equal to  $-q$ . 1
- (ii) Show that  $p = q - \frac{1}{q}$ . 2
- (c) A forklift is driving down a warehouse aisle. The acceleration of the forklift is given by the equation:
- $$\ddot{x} = -\frac{1}{2} \mu^2 e^{-x}$$
- where  $x$  is the displacement from the origin and  $\mu$  is the initial velocity at the origin.
- (i) Show that  $v^2 = 4e^{-x}$  if  $\mu = 2 \text{ ms}^{-1}$ . 1
- (ii) Explain why  $v > 0$ . 1
- (iii) Find an equation for  $x$  in terms of  $t$ . 2
- (iv) Describe the motion of the particle as  $t \rightarrow \infty$ . 1

Question 5 (12 marks) START A NEW PAGE

Marks

(a)



In the diagram, XY is a common tangent to two non-intersecting circles. This tangent touches one circle at P and the other circle at Q. AP is a chord in one circle and BQ, a chord in the other circle, is parallel to AP. AD is a straight line, cutting one circle at A and C and the other circle at B and D.

Copy the diagram onto your answer sheet.

Prove that:

- (i)  $PC \parallel QD$ . 3
- (ii) PQBC is a cyclic quadrilateral. 2

(b) The equation of the tangent to the parabola  $y = x^2$  at the point  $P(t, t^2)$  is  $y = 2tx - t^2$ .

- (i) Show that the line passing through the focus of the parabola, perpendicular to this tangent, has equation  $y = \frac{t - 2x}{4t}$ . 2
- (ii) Show that the foot of the perpendicular from the focus to the tangent is the point  $F\left(\frac{t}{2}, 0\right)$ . 2
- (iii) Find the locus of M, the midpoint of PF. 3

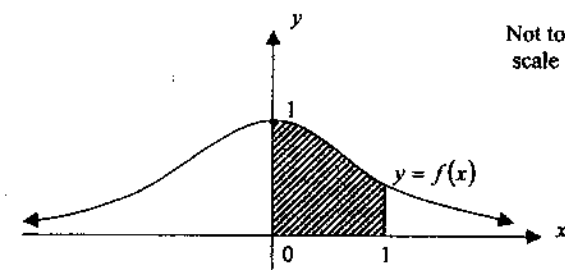
Question 6 (12 marks) START A NEW PAGE

Marks

(a) A crew of four rowers is to be chosen from five boys and six girls. How many different crews are possible if:

- (i) there are no restrictions? 1
- (ii) the shortest girl and the tallest boy must be included? 1

(b) Consider the graph of the function  $f(x) = \frac{1}{1+x^2}$ .



- (i) Find the area bounded by this curve, the x-axis and the two ordinates  $x = 0$  and  $x = 1$  using Simpson's Rule with three function values. Answer correct to 4 decimal places. 2
- (ii) Find the exact value of the area bounded by  $y = f(x)$ , the x-axis and the two ordinates  $x = 0$  and  $x = 1$ . 2
- (iii) Hence find an approximation for  $\pi$  correct to 2 decimal places. 1
- (c) Surveyors have marked out two points, A and B, in St Peter's St. The points are 52m apart and B is due east of A. 5

The bearings of A and B from the tallest point of the Great Hall are  $230^\circ$  T and  $110^\circ$  T respectively. The angles of elevation of the tallest point of the Great Hall from A and B are  $30^\circ$  and  $60^\circ$  respectively.

Show that the tallest point of the Great Hall is  $4\sqrt{39}$  m high.

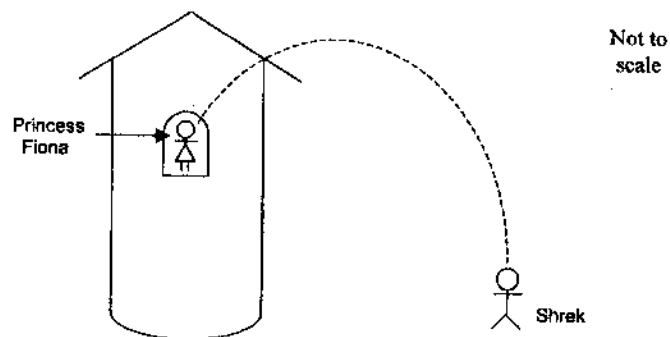
Question 7 (12 marks) START A NEW PAGE

Marks

- (a) Find all the values of  $\theta$  for which  $\cos^2 \theta + \frac{\sqrt{3}}{2} \sin 2\theta = 0$ .

4

(b)



Princess Fiona is locked up in a tower, 80m above the ground. To gain the attention of Shrek, Princess Fiona throws a lentil at an angle of elevation of  $\theta$  and an initial velocity of  $50\text{ms}^{-1}$ .

- (i) Derive the equations for the horizontal and vertical displacements of the lentil  $t$  seconds after it is thrown. (Use  $g \approx 10\text{ms}^{-2}$ .)
- (ii) Shrek is 300m from the base of the tower when he is hit by the lentil. Find the values of the initial angle of projection,  $\theta$ , correct to the nearest degree, if Shrek is 2m tall.

4

4

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End of Paper

MATHEMATICS EXT 1  
TRIAL EXAM, 2004  
SOLUTIONS

QUESTION 1: (12 marks) Calc 1/6

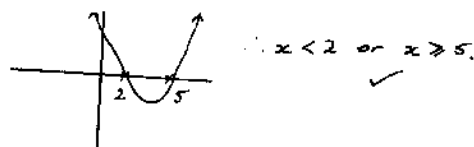
(a)  $\frac{3}{x-2} \leq 1, x \neq 2$  ✓

$3(x-2) \leq (x-2)^2$  ✓

$3x-6 \leq x^2-4x+4$

$0 \leq x^2-7x+10$

$0 \leq (x-5)(x-2)$



(b)  $m_1 = 4$  and  $m_2 = -\frac{3}{2}$

$\tan \theta = \left| \frac{4 - (-\frac{3}{2})}{1 + 4(-\frac{3}{2})} \right|$  ✓

$= \frac{11}{10}$

$\therefore \theta = 47^\circ 44'$  ✓

(c)  $\lim_{x \rightarrow 0} \frac{\sin 4x}{8x} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{\sin 4x}{4x}$   
 $= \frac{1}{2} \times 1$   
 $= \frac{1}{2}$  ✓

(d)  $\int_0^{\pi/3} \sin^2 3x \, dx$   
 $= \frac{1}{2} \int_0^{\pi/3} 1 - \cos 6x \, dx$  ✓  
 $= \frac{1}{2} \left[ x - \frac{1}{6} \sin 6x \right]_0^{\pi/3}$  ✓

$= \frac{1}{2} \left[ \left( \frac{\pi}{3} - 0 \right) - 0 \right]$

$= \frac{\pi}{6}$  ✓

Calc 3

(e)  $\int_0^1 x(1-x)^7 \, dx$

$u = 1-x$

$\frac{du}{dx} = -1$

when  $x=0, u=1$

$x=1, u=0$

$= \int_1^0 (1-u) \cdot u^7 \cdot -du$  ✓

$= \int_0^1 u^7 - u^8 \, du$

$= \left[ \frac{u^8}{8} - \frac{u^9}{9} \right]_0^1$  ✓

$= \frac{1}{8} - \frac{1}{9}$

$= \frac{1}{72}$  ✓

Calc 3

Comments:

(a) Must state that  $x \neq 2$ .

(b) Learn formula correctly - complete with absolute value signs!  
Be careful with minus sign too.

(c) ✓

(d) Many incorrect substitutions for  $\sin^2 3x$ .

(e) Show all working  
Don't forget to change the limits  
NB  $\int_0^1 f(x) \, dx \neq \int_1^0 f(x) \, dx$ .

QUESTION 2: (12 marks) Com 1  
Reas 3

(a)  $y = x^2 \cdot \sin^{-1} 3x$

$u = x^2 \quad v = \sin^{-1} 3x$

$u' = 2x \quad v' = \frac{3}{\sqrt{1-9x^2}}$

$\frac{dy}{dx} = 2x \sin^{-1}(3x) + \frac{3x^2}{\sqrt{1-9x^2}}$  ✓

(b) PARABOLA

No. of arrangements  $= \frac{8!}{3!}$  ✓  
 $(= 6720)$

(c)  $P(x) = ax^3 - 8x^2 - 9$

If divisible by  $x-a$ , then  $P(a) = 0$

$0 = a^3 - 8a^2 - 9$  ✓

$0 = (a^2 - 9)(a^2 + 1)$

$\therefore$  Since  $a$  is real,  $a = \pm 3$ . ✓

(d)  $y = \cos^{-1} x$  or  $y = 2\cos^{-1}(1-x)$

$\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$

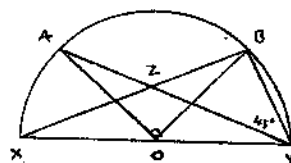
$\frac{dy}{dx} = -\frac{2}{1+(1-x)^2}$

when  $x=0, \frac{dy}{dx} = -1$  when  $x=0, \frac{dy}{dx} = -1$  ✓

$y - \pi/2 = -1(x-0)$

$x+y - \pi/2 = 0$  ✓

(e)



(i)  $\angle AYB = 45^\circ$  because the angle at the centre is twice the angle at the circumference, standing on the same arc, AB. ✓

(ii) Also,  $\angle XBY = 90^\circ$  ( $\angle$  in a semicircle is  $90^\circ$ ) ✓

$\therefore \angle BZY = 45^\circ$  ( $\angle$  sum  $\Delta = 180^\circ$ ) ✓

$\therefore BY = BZ$  (sides opposite = angles in a  $\Delta$  are  $=$ ). ✓

Reas 3

Comments:

a) to differentiate  $\sin^{-1} f(x)$  it is more successful to use the rule.

$\frac{d}{dx} (\sin^{-1} f(x)) = \frac{1}{\sqrt{1-f(x)^2}} \cdot f'(x)$

b) Well done.

c) MUST BE stated that  $P(a) = 0$

The resulting equation is a quadratic. It was solved very badly. You should recognise equations of this form.

d) Really only need to find one tangent gradient because it is a common tangent.

e) Word of advice!

Draw a clear/large diagram. Mark on everything you can find. The solution generally reveals itself.

QUESTION 3: (12 marks) Com 5

(a) (i)  $\sqrt{3} \cos x - \sin x$

$R \cos(x + \alpha) = R \cos x \cos \alpha - R \sin x \sin \alpha$

$R \cos \alpha = \sqrt{3}$

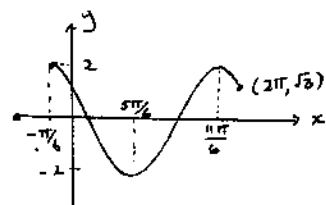
$R \sin \alpha = 1$

$R = 2$  and  $\tan \alpha = \frac{1}{\sqrt{3}}$

$\alpha = \pi/6$

$\therefore \sqrt{3} \cos x - \sin x = 2 \cos(x + \pi/6)$

(ii)



Com 1

(iii)  $2 \cos(x + \pi/6) = \sqrt{2}$

$\cos(x + \pi/6) = \frac{1}{\sqrt{2}}$

$\therefore x + \pi/6 = \frac{\pi}{4}, \frac{7\pi}{4}$

$\therefore x = \frac{\pi}{12}, \frac{19\pi}{12}$

(b) (i)  $x = a \cos(4t + \alpha)$

$\dot{x} = -4a \sin(4t + \alpha)$

$\ddot{x} = -16a \cos(4t + \alpha)$

$= -16x$  as required

(ii)  $x = 5, t = 0 \Rightarrow 5 = a \cos \alpha$  ①

$v = -4, t = 0 \Rightarrow -4 = -4a \sin \alpha$

$1 = a \sin \alpha$  ②

① + ②  $\Rightarrow 25 + 1 = a^2$

$a = 26$

$a = \sqrt{26}$

(iii) Maximum speed is  $4\sqrt{26}$  cm/s

(c)

$n=0: 5^0 + 11 = 1 + 11 = 12$   
which is divisible by 4

Assume true for  $n=k$ :

ie  $5^k + 11 = 4M$  for some integer  $M$ .

Investigate  $n=k+1$ :

$5^{k+1} + 11 = 5 \cdot 5^k + 11$   
 $= 5(4M - 11) + 11$   
using assumption

$= 20M - 44$

$= 4(5M - 11)$

$= 4P, (P \in \mathbb{Z})$

$\therefore$  If proposition true for  $n=k$ , it

is also true for  $n=k+1$ .

Since it is true for  $n=0$ , it is also

true for  $n=1, 2, \dots$  and hence

all positive integers by the principle of mathematical induction.

Com 3

(a) (ii) mark the endpoints on your curve and make sure it was greater than 1 cycle of the curve.

(iii) Don't forget answer in all appropriate quadrants.

(b) (i) careful with derivative.  $\frac{d}{dx}(\cos x) = -\sin x$

(ii) poorly done. Many algebraic errors.

(iii) poor.

(c) NB Initial value is  $n=0$ !

Also careless e.g.  $5 \cdot 5^k + 11 \neq 5(5^k + 11)$

QUESTION 4: (12 marks)

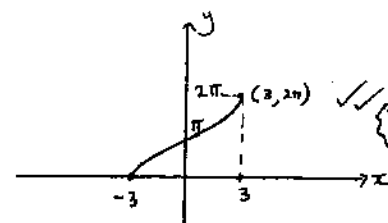
Com 5  
New Calc 3/4

(a)  $f(x) = \pi + 2 \sin^{-1}(\frac{x}{3})$

(i) Domain:  $-3 \leq x \leq 3$

Range:  $0 \leq f(x) \leq 2\pi$

(ii)



Com 2

(b)  $x^3 + px^2 + q = 0$

(i) Let roots be  $\alpha, \frac{1}{\alpha}$  and  $\beta$ .

$\therefore$  Product of roots:

$\alpha \cdot \frac{1}{\alpha} \cdot \beta = -q$

$\therefore \beta = -q$

$\therefore$  The third root is  $-q$ .

(ii)  $\Sigma$  of roots:

$\alpha + \frac{1}{\alpha} - q = -p$  ①

$\Sigma$  of pairs of roots:

$\alpha \cdot \frac{1}{\alpha} - \alpha q - \frac{q}{\alpha} = 0$

$1 - q(\alpha + \frac{1}{\alpha}) = 0$

but from ①:  $\alpha + \frac{1}{\alpha} = q - p$

$\therefore 1 - q(q - p) = 0$

$1 - q^2 + pq = 0$

$\therefore pq = q^2 - 1$

$p = q - \frac{1}{q}$

Real 3

(c)  $\ddot{x} = -\frac{1}{2}\mu^2 e^{-x}$  where  $\mu = 2$

i)  $\dot{x} = \frac{d}{dx}(\frac{1}{2}v^2)$

$\frac{d}{dx}(\frac{1}{2}v^2) = -\frac{1}{2} \cdot 2^2 \cdot e^{-x}$

$\frac{1}{2}v^2 = \int -2e^{-x} dx$

$\frac{1}{2}v^2 = 2e^{-x} + C$

when  $x=0, v=2$

$\frac{1}{2} \cdot 4 = 2e^0 + C$

$2 = 2 + C$

$C = 0$

$\frac{1}{2}v^2 = 2e^{-x}$

$v^2 = 4e^{-x}$

(Calc1)

ii)  $v = \pm \sqrt{4e^{-x}}$

$= \pm 2e^{-x/2}$

Since  $e^{-x/2} > 0$  for all  $x$  and the initial conditions gives the velocity is 2 m/s. (positive velocity)

$\therefore v > 0$  for all  $x$

$v = 2e^{-x/2}$

(Comm1)

iii)  $\frac{dx}{dt} = 2e^{-x/2}$

$\frac{dx}{dx} = \frac{1}{2}e^{x/2}$

$t = \int \frac{1}{2}e^{x/2} dx$

$t = \frac{1}{2} \times \frac{1}{\frac{1}{2}} e^{x/2} + C$

$t = e^{x/2} + C$

when  $t=0, x=0$

$0 = e^0 + C$

$0 = 1 + C$

$C = -1$

$t = e^{x/2} - 1$

$e^{x/2} = t + 1$

$\frac{x}{2} = \ln(t + 1)$

$x = 2 \ln(t + 1)$

(Calc3)

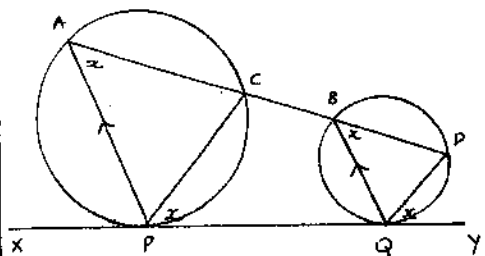
iv) As  $t \rightarrow \infty, x \rightarrow \infty$

$v \rightarrow 0$

$a \rightarrow 0$

QUESTION 5: (12 marks) Reas 12

(a)



(i) Let  $\angle CPQ = x$

$\therefore \angle PAC = x$  ( $\angle$  in the alt seg =  $\angle$  between a tangent + chord).

$\therefore \angle DBQ = x$  (corresponding  $\angle$  = as  $AP \parallel BQ$ )

$\therefore \angle DQY = x$  ( $\angle$  in alt seg =  $\angle$  between tangent + chord).

$\therefore \angle CPQ = \angle DQY$

$\therefore CP \parallel DQ$  (corresponding  $\angle$  =)

(ii)  $\therefore \angle CBQ = 180 - x$  ( $\angle$  2 str. line =  $180^\circ$ )

$PQBC$  is a cyclic quadrilateral since opposite angles are supplementary.

(b)  $y = x^2$ ,  $P(t, t^2)$

Tangent  $y = 2tx - t^2$

(i)  $m = -\frac{1}{2t}$

Focus  $(0, \frac{1}{4})$

$y - \frac{1}{4} = -\frac{1}{2t}(x - 0)$

$\therefore y = -\frac{1}{2t}x + \frac{1}{4}$

$y = \frac{t-2x}{4t}$

(ii) Solving simultaneously,

$y = 2tx - t^2$  ①

$y = \frac{t-2x}{4t}$  ②

$\therefore 2tx - t^2 = \frac{t-2x}{4t}$

$8t^2x - 4t^3 = t - 2x$

$x(8t^2 + 2) = t + 4t^3$

$x = \frac{t(1+4t^2)}{2(4t^2+1)}$   
 $= \frac{t}{2}$

$\therefore y = 7t \cdot \frac{t}{2} - t^2$   
 $= 0$

$\therefore F(\frac{t}{2}, 0)$

(iii)  $P(t, t^2)$   $F(\frac{t}{2}, 0)$

$M = (\frac{3t}{4}, \frac{t^2}{2})$

$\therefore x = \frac{3t}{4}$  and  $y = \frac{t^2}{2}$

$\therefore t = \frac{4x}{3}$

$\therefore y = \frac{1}{2} \cdot (\frac{4x}{3})^2$   
 $= \frac{8x^2}{9}$

Comments:

(a) many no. attempts.

(b) (i) Line passes thru  $S_{\text{alt}}$

(ii) Need to solve simult eqs both lines

(iv) still some careless, but improving.

Reas 12

QUESTION 6: (12 marks) Calc 2 Reas 6

(a) (i)  ${}^{11}C_4 = 330$  ✓

(ii)  ${}^9C_2 = 36$  ✓

(b) (i)  $A = \frac{1}{3} \left( 1 + 4 \cdot \frac{4}{5} + \frac{1}{5} \right)$   
 $= \frac{47}{60}$

$= 0.7833$  ✓

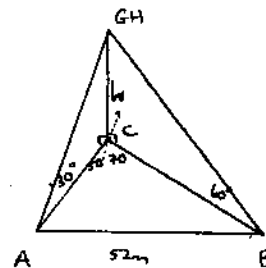
(ii)  $A = \int_0^1 \frac{1}{1+x^2} dx$  ✓

$= [\tan^{-1}x]_0^1$   
 $= \frac{\pi}{4}$  ✓ (Calc 2)

(iii)  $\therefore \frac{\pi}{4} = 0.7833$

$\therefore \pi = 3.13$  ✓ (Reas 1)

(c)



$\tan 60 = \frac{h}{BC}$

$\therefore BC = \frac{h}{\sqrt{3}}$  ✓

$\tan 30 = \frac{h}{AC}$

$\therefore AC = h\sqrt{3}$  ✓

$\therefore \cos 120 = \frac{AC^2 + BC^2 - 52^2}{2 \cdot AC \cdot BC}$  ✓

$-\frac{1}{2} = \frac{3h^2 + \frac{h^2}{3} - 52^2}{2 \cdot h\sqrt{3} \cdot \frac{h}{\sqrt{3}}}$

$-h^2 = 3h^2 + \frac{h^2}{3} - 52^2$  ✓

$\frac{13h^2}{3} = 52^2$

$\therefore h^2 = 624$

$\therefore h = \sqrt{624}$   
 $= 4\sqrt{39}$  ✓

Reas 5

Comments:

a) well done.

b) Learn Simpson's rule properly.

ii) Very easy! use the standard integral page.

iii) Hence means you must use your answers from parts i) and ii)

Draw a clear diagram. It is easier to solve this problem using the simplified expressions for BC and AC. Watch your rearranging of algebra!



QUESTION 7: (12 marks) Rev 6  
Calc 1/5

(a)  $\cos^2 \theta + \frac{\sqrt{3}}{2} \sin 2\theta = 0$

$\cos^2 \theta + \sqrt{3} \sin \theta \cos \theta = 0$  ✓

$\cos \theta (\cos \theta + \sqrt{3} \sin \theta) = 0$

$\therefore \cos \theta = 0$  or  $\cos \theta + \sqrt{3} \sin \theta = 0$

$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$   $\tan \theta = -\frac{1}{\sqrt{3}}$

$\theta = \frac{5\pi}{6}, \frac{7\pi}{6}, \dots$

$3\pi - \frac{\pi}{6}, 4\pi - \frac{\pi}{6}, \dots$

$\therefore \theta = (2n+1)\frac{\pi}{2} \quad n \in \mathbb{Z}$  ✓

$\theta = n\pi - \frac{\pi}{6}$  ✓ (Lea 4)

(b) (i)  $\ddot{x} = 0$

$x = C$

when  $t=0$ ,  $x = 50 \cos \theta \therefore C = 50 \cos \theta$

$x = 50 \cos \theta$  ✓

$x = 50t \cos \theta + C$

when  $t=0$ ,  $x = 0 \therefore C = 0$

$x = 50t \cos \theta$  ✓

$\ddot{y} = -10$

$y = -10t^2 + C$

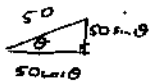
when  $t=0$ ,  $y = 50 \sin \theta \therefore C = 50 \sin \theta$

$y = -10t^2 + 50 \sin \theta$  ✓

$y = -5t^2 + 50t \sin \theta + C$

when  $t=0$ ,  $y = 80 \therefore C = 80$

$y = -5t^2 + 50t \sin \theta + 80$  ✓



(ii) when  $x = 300$ ,  $y = 2$ .

$300 = 50t \cos \theta$

$\therefore t = \frac{6}{\cos \theta}$  ✓

$2 = -5 \cdot \frac{36}{\cos^2 \theta} + 50 \cdot \frac{6}{\cos \theta} \sin \theta + 80$  ✓

$0 = -180 \sec^2 \theta + 300 \tan \theta + 78$

$= -180 (\tan^2 \theta + 1) + 300 \tan \theta + 78$

$= -180 \tan^2 \theta + 300 \tan \theta - 102$

$0 = 180 \tan^2 \theta - 300 \tan \theta + 102$  ✓

$\therefore \tan \theta = \frac{300 \pm \sqrt{300^2 - 4 \cdot 180 \cdot 102}}{2 \cdot 180}$

$= 1.19 \quad \text{or} \quad 0.475$

$\therefore \theta = 49^\circ 58' \quad \text{or} \quad 25^\circ 27'$  ✓

$\therefore$  The initial angle of projection could be  $50^\circ$  or  $25^\circ$  to the nearest degree.

Calc  
1/5

Comments:

(a) Factorise!!

"All solutions" means find the general solution.  
really should indicate that  $n$  is an integer

(b) (i) 'derive' means you must show all steps, NOT just quote a formula.

(ii) finding a  $t$  value first wastes far too much time. Eliminate  $t$  and find the angles straight away.